

Sentiment Analysis of Rome Colosseum Visitor Reviews

Rome Colosseum Visitor Reviews Dataset (2019 - 2026)
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1. Problem Statement & Dataset Summary

This project builds a complete sentiment analysis pipeline using real visitor review data for the Rome Colosseum, one of the most visited landmarks in the world.

Objective: Classify visitor reviews into Positive, Neutral, or Negative sentiment, with special attention to detecting minority-class negative reviews, since these matter most for tourism management follow-up.

Attribute	Detail
Total reviews	8,285 (after removing duplicates)
Time range	May 2019 - June 2026
Key columns	text (review), rating (1-5 stars), sentiment_label
Other features	tripType, travel_season, word_count, published_platform
Missing values	Minimal (no nulls in key columns)

2. Exploratory Data Analysis (EDA)

2.1 Star Rating Distribution

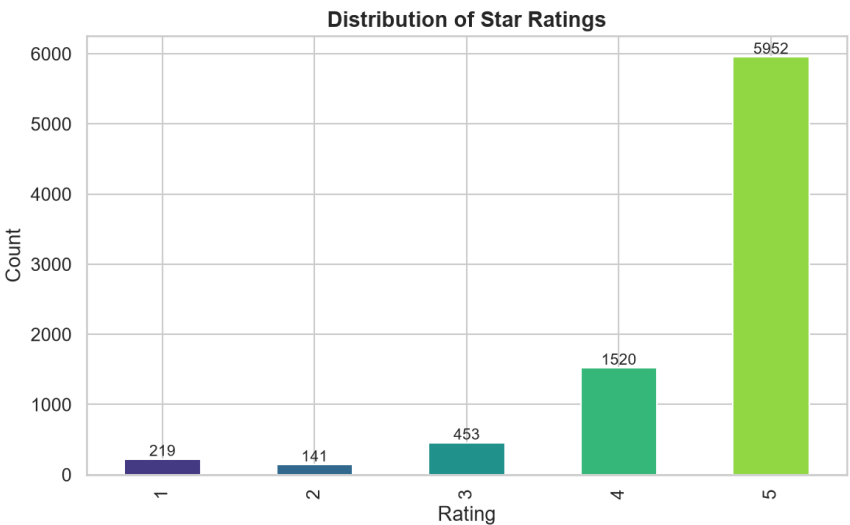


Figure 1: Distribution of star ratings across all reviews.

The vast majority of reviews are 5-star (over 60%), followed by 4-star. Only around 4% of reviews are 1 or 2 stars, confirming the heavy positive skew typical of tourist attraction reviews.

2.2 Sentiment Label Comparison

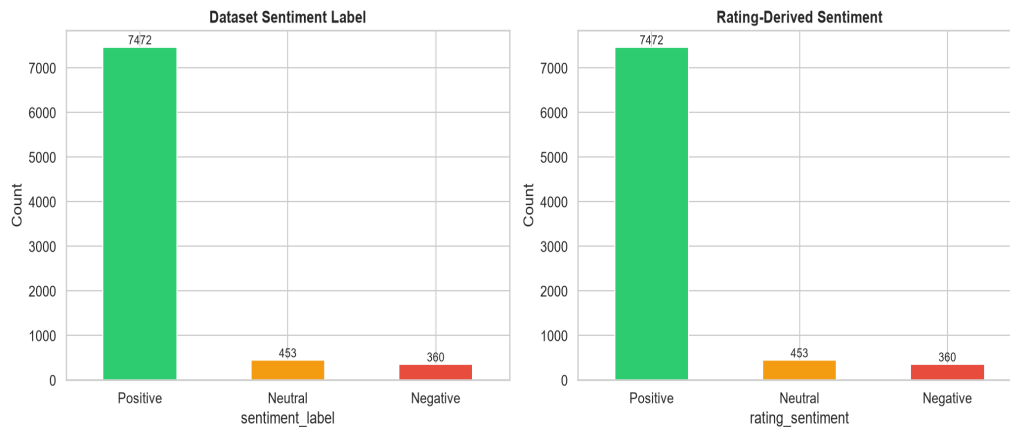


Figure 2: Dataset sentiment label vs. rating-derived sentiment.

The dataset's own sentiment_label agrees with rating-derived sentiment (4-5 stars = Positive, 3 stars = Neutral, 1-2 stars = Negative) at over 95% agreement. Class distribution: Positive = 7,472 (90.2%), Neutral = 453 (5.5%), Negative = 360 (4.3%).

2.3 Trip Type, Season & Review Length

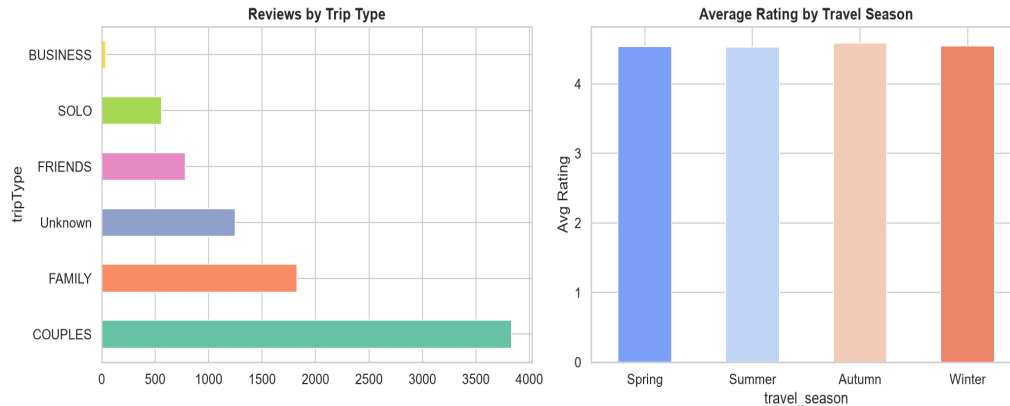


Figure 3: Review volume by trip type (left) and average rating by season (right).

Couples dominate trip types, followed by Family and Friends. Spring and Summer show the highest review volumes, and average ratings stay stable across all seasons at around 4.5 stars.

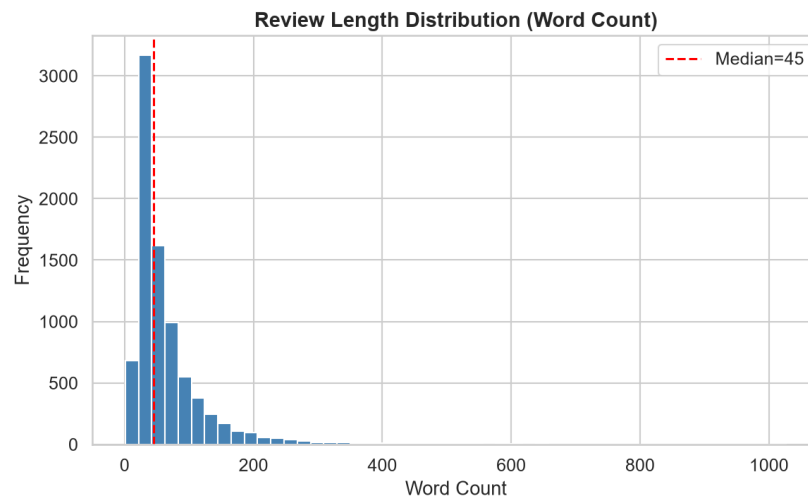


Figure 4: Distribution of review length in words (median = 45).

Most reviews fall between 25 and 100 words. A small number of outliers exceed 400 words, usually detailed trip reports.

2.4 Key Observations

- The dataset is heavily imbalanced: about 90% of reviews are Positive, making class imbalance the central challenge.
- Rating-derived sentiment closely matches the dataset's sentiment labels (over 95% agreement).
- Couples is the dominant trip type; Spring and Summer are the peak review seasons.
- Average ratings remain stable across seasons, suggesting consistent visitor satisfaction year-round.
- Most reviews are medium length (25-75 words); very long reviews (200+ words) are rare but carry richer sentiment signal.

3. Text Preprocessing

Pipeline steps:

1. Lowercasing - normalize all text
2. URL removal - strip web links
3. Punctuation/number removal - keep only alphabetic characters
4. Tokenization - split into individual words (NLTK word_tokenize)
5. Stopword removal - remove common English stopwords
6. Lemmatization - reduce words to base form (NLTK WordNetLemmatizer)

Key Decision: Keeping Negation Words

Negation words such as 'not', 'never', 'don't', 'can't', and 'won't' are deliberately kept during stopword removal. These words flip sentiment polarity - removing them would turn 'not good' into 'good', completely changing the meaning. This matters a lot for accurate sentiment classification.

Before / After Examples

#	Before (raw)	After (cleaned)
1	A must see for any visitor to Rome, but go to the visitor centre	must see visitor rome visitor centre buy front queue pass ticket...
2	We booked a tour... horrific experience... the tour guide never showed up	booked tourhorrific experience tour guide never showed...
3	Not worth the (euro)30 entry ticket. Best to take a picture from outside.	not worth entry ticket best take picture outside...

4. Lexicon-Based Baseline (VADER)

VADER (Valence Aware Dictionary and sEntiment Reasoner) was applied directly to the raw review text. Compound scores were mapped to labels as: score of 0.05 or above = Positive, -0.05 or below = Negative, and everything in between = Neutral.

Metric	Score
Macro F1	0.4505
Positive Recall	0.906
Negative Recall	0.603
Neutral Recall	0.049

VADER performs reasonably on Positive and Negative reviews but fails badly on Neutral reviews. This is expected, since VADER struggles with ambivalent or mixed-tone text.

5. ML Model Comparison & Class Imbalance

5.1 Why Accuracy is Misleading

With about 90% of reviews being Positive, a dummy classifier that always predicts 'Positive' would score around 90% accuracy, but would have 0% recall on Negative and Neutral reviews. That is not useful for the actual business goal, which is catching negative feedback. We therefore use macro-averaged F1 as the primary metric, since it treats all three classes equally regardless of their size.

5.2 Results Comparison Table

Model	Macro F1	Kappa	Bal. Acc	Pos P	Pos R	Neu P	Neu R	Neg P	Neg R
LR (baseline)	0.382	0.100	0.370	0.908	0.999	0.000	0.000	0.800	0.111
MNB (baseline)	0.316	0.000	0.333	0.902	1.000	0.000	0.000	0.000	0.000

Model	Macro F1	Kappa	Bal. Acc	Pos P	Pos R	Neu P	Neu R	Neg P	Neg R
LR (balanced weight)	0.581	0.395	0.621	0.958	0.912	0.215	0.341	0.489	0.611
LinearSVC (balanced)	0.535	0.342	0.503	0.932	0.974	0.255	0.132	0.592	0.403
LR (undersampled)	0.497	0.258	0.673	0.980	0.727	0.138	0.582	0.311	0.708
LR (oversampled)	0.587	0.410	0.601	0.951	0.937	0.255	0.297	0.519	0.569
LR (tuned+balanced)	0.562	0.381	0.556	0.944	0.950	0.250	0.231	0.515	0.486

Best two configurations shown in bold (LR balanced weight, LR oversampled).

5.3 Model Comparison Chart

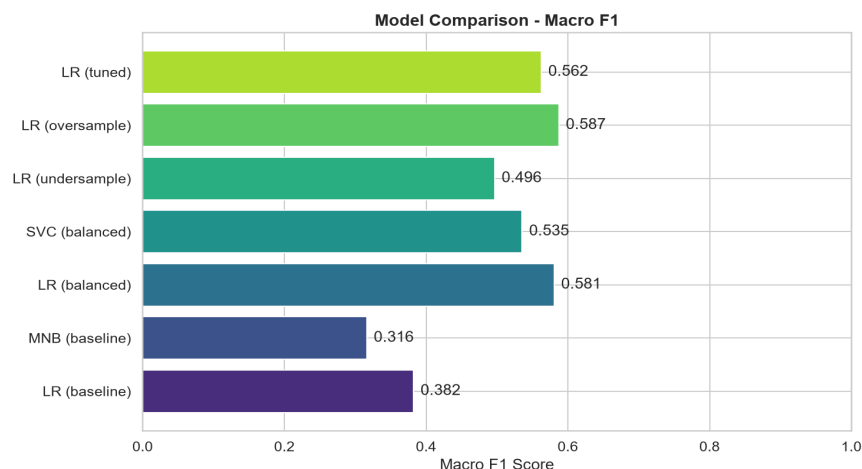


Figure 5: Macro F1 score across all models and imbalance-handling techniques.

5.4 Techniques Comparison

- Baseline models (LR, MNB) with no imbalance handling reach high accuracy (~90%) but have zero recall on minority classes - completely useless for flagging negative reviews.
- Class weighting (class_weight='balanced') is the simplest fix and delivers the best Negative recall (0.611) with strong macro F1.
- Random oversampling achieves the highest overall macro F1 (0.587) with fairly balanced performance across classes.
- Random undersampling maximizes Negative recall (0.708) but sacrifices too much Positive precision, so it throws away useful data.

5.5 Trade-off Recommendation

For a tourism company that wants to flag negative reviews for follow-up, Logistic Regression with class_weight='balanced' is recommended because it gives the highest Negative recall (0.611), meaning it catches 61% of genuinely negative reviews, while keeping a good macro F1 (0.581) and needing no data duplication or loss. The small drop in Positive precision (0.958 vs 0.908 for the baseline) is an acceptable trade-off given the business goal.

6. Error Analysis

Out of 1,657 test reviews, 181 were misclassified (89.1% accuracy). Key error patterns are summarized below.

Mixed sentiment: praises the Colosseum but complains about queues/scammers — *why it fails:* model latches onto positive words and misses the complaint.

Short reviews: "Course stay rome mythical monument" — *why it fails:* too little context for a reliable prediction.

Sarcasm / irony: "Amazing wait times" (meant negatively) — *why it fails:* word-level models read 'amazing' as positive.

Neutral ambiguity: 3-star reviews with both praise and criticism — *why it fails:* sits right on the decision boundary.

Domain-specific complaints: queues, heat, ticket issues, scams — *why it fails:* uses fairly neutral vocabulary the model doesn't flag.

7. Conclusion

Recommended Approach

Logistic Regression with `class_weight='balanced'`, using TF-IDF features (unigrams + bigrams). It gives the best balance of simplicity, interpretability, and minority-class detection - which is what matters most for real-world tourism feedback management.

One Limitation

The Neutral class remains poorly predicted across every model tested (best F1 is only about 0.264). This happens because neutral reviews mix positive and negative language, which overlaps heavily with the other two classes in TF-IDF feature space.

One Improvement Idea

Fine-tune a pre-trained transformer model such as BERT or DistilBERT on this dataset. Transformers capture context, sarcasm, and negation far better than bag-of-words models, and would likely improve both Neutral and Negative classification.

Report generated as part of the Summer AI Internship 2026 assignment.